

Requirement Analysis Report

E-Commerce Order Management Database System — Week 1

1. Purpose of This Document

This report presents the requirement analysis carried out for the E-Commerce Order Management Database System. It documents the business scenario, the purpose of each core entity, the major business processes, and the analysis approach used before proceeding to database design, ER modeling, normalization, and SQL implementation.

2. Business Scenario Overview

A rapidly growing e-commerce company needs a centralized database system to manage its daily operations. The system must support customer registration, product management, inventory tracking, order processing, payment management, shipment tracking, and customer reviews. The goal is to store accurate information, reduce data redundancy, improve operational efficiency, and generate meaningful business reports for decision-making.

3. Purpose of Each Core Entity

Customer

Stores the personal and contact details of every registered buyer (Customer ID, Name, Email, Mobile Number, Address, Registration Date). It is the anchor entity that links a person to the orders and reviews they create, and enables the business to communicate with and identify buyers.

Category

Groups products into logical classifications (Category ID, Category Name, Description). It allows the catalog to be organized, supports browsing/filtering, and enables category-level reporting.

Product

Represents every item offered for sale (Product ID, Product Name, Price, Stock Quantity, Category ID). It is central to inventory control, pricing, and order fulfilment, and it links each item back to its Category.

Supplier

Records the vendors who supply products (Supplier ID, Supplier Name, Contact Information). It supports procurement and restocking decisions and provides accountability for product sourcing.

Order

Captures each purchase transaction placed by a customer (Order ID, Customer ID, Order Date, Total Amount, Order Status). It is the core transactional entity that ties a customer to the products purchased, the payment made, and the shipment sent.

Order Details

Breaks an Order down into its individual line items (Order Detail ID, Order ID, Product ID, Quantity, Unit Price). It resolves the many-to-many relationship between Orders and Products and enables accurate calculation of order totals.

Payment

Tracks how and when an order was paid for (Payment ID, Order ID, Payment Method, Payment Date, Payment Status). It supports financial reconciliation, revenue reporting, and fraud/dispute handling.

Shipment

Tracks the delivery of an order (Shipment ID, Order ID, Delivery Address, Shipment Date, Delivery Status). It provides visibility into logistics performance and lets both staff and customers track delivery progress.

Review

Stores customer feedback on purchased products (Review ID, Customer ID, Product ID, Rating, Comments). It supports product quality monitoring, builds customer trust, and feeds analytics on customer satisfaction.

4. Major Business Processes

- **Customer Registration & Profile Management** — capturing and maintaining customer details.
- **Product & Category Management** — adding, updating, and classifying products in the catalog.
- **Inventory Management** — tracking stock quantity and triggering replenishment via suppliers.
- **Order Processing** — creating an order, adding order line items, and updating order status.
- **Payment Processing** — recording the payment method, date, and status for each order.
- **Shipment & Delivery Tracking** — recording delivery address, dispatch date, and delivery status.
- **Customer Review Management** — capturing ratings and comments after product purchase.
- **Reporting & Analytics** — generating sales, inventory, and customer insight reports.

5. Analysis Approach

The analysis followed a structured top-down approach: (1) study the business scenario, (2) confirm the purpose of each mandatory core entity, (3) trace the business processes that generate and consume data for each entity, (4) identify the stakeholders who participate in or depend on each process, and (5) translate the findings into functional and non-functional requirements that will drive the subsequent ER modeling and database design phases.

6. Conclusion

The E-Commerce Order Management System is a transaction-heavy platform whose success depends on accurate, non-redundant data across nine interlinked core entities. This report confirms that the predefined entity set is sufficient to model the full order lifecycle — from customer registration through product selection, ordering, payment, shipment, and post-purchase review — and forms the basis for the Business Requirement Document and SRS that follow.